

MP363 – Quantum Mechanics I

Problem Set 2

Due by the end of lecture on Friday, 6 November 2015

1. An inner product on a vector space is any object that takes two vectors u and v and assigns to them a complex number $\langle u | v \rangle$ which satisfies the following for any vectors u, v and w and any complex number λ :

$$(i) \quad \langle u | v + w \rangle = \langle u | v \rangle + \langle u | w \rangle$$

$$(ii) \quad \langle u + v | w \rangle = \langle u | w \rangle + \langle v | w \rangle$$

$$(iii) \quad \langle u | \lambda v \rangle = \lambda \langle u | v \rangle$$

$$(iv) \quad \langle \lambda u | v \rangle = \lambda^* \langle u | v \rangle$$

$$(v) \quad \langle u | v \rangle^* = \langle v | u \rangle$$

- (a) If the vector space is the space of column vectors of the form

$$v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

where $v_i \in \mathbb{C}$, then show that

$$\langle u | v \rangle = \sum_{i=1}^n u_i^* v_i$$

satisfies all the above identities (and thus defines an inner product).

- (b) If the vector space is the space of complex-valued functions on the real line, then show that

$$\langle u | v \rangle = \int_{-\infty}^{\infty} u^*(x) v(x) dx$$

also satisfies all the above identities.

2. If A is a complex $n \times n$ matrix, its adjoint $A^\dagger$ is given by

$$A^\dagger = (A^T)^* = (A^*)^T.$$

Use this to prove the following for any matrices A and B and any complex number α :

- (a) $(A + B)^\dagger = A^\dagger + B^\dagger$
- (b) $(\alpha A)^\dagger = \alpha^* A^\dagger$
- (c) $(AB)^\dagger = B^\dagger A^\dagger$
- (d) $(A^\dagger)^\dagger = A$

Note: Even though you proved these properties for $n \times n$ matrices, they hold for *any* linear operators on *any* Hilbert space. For example, even if A and B are differential operators acting on functions, $(AB)^\dagger = B^\dagger A^\dagger$ still holds.

3. An inner product on the vector space of 2×2 complex matrices is given by

$$\langle M | N \rangle = \frac{1}{2} \text{tr} (M^\dagger N)$$

where tr is the trace operation.

- (a) For the matrices

$$M = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 2+i & 1 \\ -1 & i \end{pmatrix},$$

compute $\langle M | N \rangle$ and $\langle N | M \rangle$ and confirm that they are complex conjugates of one another.

- (b) Show that each of the three matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are self-adjoint and they form an orthonormal set of matrices, i.e. $\langle \sigma_i | \sigma_j \rangle = \delta_{ij}$.

(Note: This inner product satisfies condition (v) from Problem 1, so you do not have to compute all nine possibilities. For example, if you show that $\langle \sigma_2 | \sigma_3 \rangle = 0$, then it follows automatically that $\langle \sigma_3 | \sigma_2 \rangle$ vanishes as well.)

4. In lecture, we showed that multiplication by any real function is a self-adjoint operation, the momentum operator $\hat{P}$ is self-adjoint and that any power of a self-adjoint operator is also self-adjoint. These facts lead us to conclude that the Hamiltonian is self-adjoint.

We can also show this another (longer) way: for simplicity, suppose we only have a single spatial dimension x , so that the inner product between two wavefunctions (both with finite norm) is

$$\langle \psi | \zeta \rangle = \int_{-\infty}^{\infty} \psi^*(t, x) \zeta(t, x) dx$$

and the Hamiltonian is

$$H = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(t, x)$$

where the potential V is a real function.

- (a) To set up the proof that H is self-adjoint, first show that for any two functions f and g ,

$$f \frac{\partial^2 g}{\partial x^2} - \frac{\partial^2 f}{\partial x^2} g = \frac{\partial}{\partial x} \left(f \frac{\partial g}{\partial x} - \frac{\partial f}{\partial x} g \right).$$

- (b) Use what you just proved in (a) to show that

$$\langle \psi | H \zeta \rangle - \langle H \psi | \zeta \rangle = 0$$

for all ψ and ζ , and thus H is self-adjoint.